Chapter 1. Hello, world of concurrency in C++!

This chapter covers

- What is meant by concurrency and multithreading
- Why you might want to use concurrency and multithreading in your applications
- Some of the history of the support for concurrency in C++
- What a simple multithreaded C++ program looks like

These are exciting times for C++ users. Thirteen years after the original C++ Standard was published in 1998, the C++ Standards Committee is giving the language and its supporting library a major overhaul. The new C++ Standard (referred to as C++11 or C++0x) was published in 2011 and brings with it a whole swathe of changes that will make working with C++ easier and more productive.

One of the most significant new features in the C++11 Standard is the support of multithreaded programs. For the first time, the C++ Standard will acknowledge the existence of multithreaded applications in the language and provide components in the library for writing multithreaded applications. This will make it possible to write multithreaded C++ programs without relying on platform-specific extensions and thus allow writing portable multithreaded code with guaranteed behavior. It also comes at a time when programmers are increasingly looking to concurrency in general, and multithreaded programming in particular, to improve application performance.

This book is about writing programs in C++ using multiple threads for concurrency and the C++ language features and library facilities that make that possible. I'll start by explaining what I mean by concurrency and multithreading and why you would want to use concurrency in your applications. After a quick detour into why you might *not* want to use it in